RBNS FunctionalArchitecture v1.0

Requirements Traceability Matrix

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Coverage Summary

31 REQUIREMENTS	31 COVERED	0 UNCOVERED	100.0% COVERAGE	0 ALL PASS	0 HAS FAILURES

Traceability Matrix

Req ID	Type	Title	Description	Source	Parent	Test Cases	Risk Controls	Coverage
UN-001	functional	**Physician navigation to target nodule**	Physicians need to navigate a bronchoscope to a target nodule identified in preoperative imaging in order to obtain a tissue biopsy.	—	—	**TC-001**: not_run	—	covered
UN-002	functional	**Peripheral airway reach**	Physicians need to reach target nodules located in peripheral (small-diameter, high-generation) airway branches that are difficult or impossible to access with manual bronchoscopy.	—	—	**TC-002**: not_run	—	covered
UN-003	safety	**No unintended airway injury**	Patients need assurance that navigation will not cause unintended injury to the airway.	—	—	**TC-003**: not_run	—	covered
UN-004	functional	**Physician supervisory authority**	Physicians need to retain supervisory authority over the procedure including the ability to pause or abort navigation at any time.	—	—	**TC-004**: not_run	—	covered
UN-005	functional	**Procedure record**	Clinical staff need a record of the navigation procedure to support post-procedure review.	—	—	**TC-005**: not_run	—	covered

Req ID	Type	Title	Description	Source	Parent	Test Cases	Risk Controls	Coverage
SYS-REQ-001	functional	**System navigation capability**	The system shall enable a physician to navigate a flexible bronchoscope to target pulmonary nodules identified in preoperative imaging.	—	UN-001	**TC-006**: not_run	—	covered
SYS-REQ-002	functional	**Positional confidence for biopsy**	The system shall provide the physician with sufficient positional confidence to perform a tissue biopsy at the target location.	—	UN-001	**TC-007**: not_run	—	covered
SYS-REQ-003	safety	**Prevent unintended scope motion**	The system shall prevent unintended scope advancement or steering that could result in airway wall injury.	—	UN-003	**TC-008**: not_run	—	covered
SYS-REQ-004	functional	**Respond to pause and abort commands**	The system shall respond to physician pause and abort commands within a clinically acceptable time.	—	UN-004	**TC-009**: not_run	—	covered
SYS-REQ-005	functional	**Generate procedure record**	The system shall generate a complete procedure record suitable for post-procedure clinical review.	—	UN-005	**TC-010**: not_run	—	covered
SYS-REQ-006	safety	**Detect and respond to navigation accuracy loss**	The system shall detect conditions under which navigation accuracy cannot be assured and shall prevent scope motion under those conditions.	—	UN-003	**TC-011**: not_run	—	covered
SUB-REQ-001	performance	**30 Hz fused pose estimate**	The RBNS shall produce a fused bronchoscope tip pose estimate at a rate of no less than 30 Hz during the NAVIGATE phase.	—	SYS-REQ-001	**TC-012**: not_run	—	covered
SUB-REQ-002	functional	**Uncertainty ellipsoid on IF-01**	Each fused tip pose estimate shall include a 3D position uncertainty representation expressed as an ellipsoid with semi-axis lengths in millimeters.	—	SYS-REQ-002	**TC-013**: not_run	—	covered
SUB-REQ-003	performance	**CBCT registration within 4 seconds**	The RBNS shall complete image-to-patient registration within 4 seconds of receiving a Cone Beam Computed Tomography (CBCT) volume.	—	SYS-REQ-002	**TC-014**: not_run	—	covered
SUB-REQ-004	safety	**Replanning inhibition under uncertainty**	The RBNS shall inhibit path replanning when the tip position uncertainty exceeds the configured replanning threshold.	—	SYS-REQ-003	**TC-015**: not_run	—	covered

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SUB-REQ-005	functional	**Configurable replanning threshold**	The replanning uncertainty threshold shall be a configurable parameter modifiable without software rebuild.	—	SYS-REQ-003	**TC-016**: not_run	—	covered
SUB-REQ-006	safety	**HOLD within one guidance cycle of uncertainty exceedance**	The RBNS shall command zero scope advancement (HOLD) within one guidance cycle of the tip position uncertainty exceeding the configured threshold.	—	SYS-REQ-003	**TC-017**: not_run	—	covered
SUB-REQ-007	safety	**Guidance command magnitude limiting**	All guidance commands shall be magnitude-limited to configurable maximum advance rate and maximum steering angle values prior to transmission on IF-07.	—	SYS-REQ-003	**TC-018**: not_run	—	covered
SUB-REQ-008	functional	**RBNS procedure state machine**	The RBNS shall implement the procedure state machine with states IDLE NAVIGATE CONFIRM BIOPSY FAULT and ABORT with transitions as defined in the RBNS procedure state machine model (D7).	—	SYS-REQ-004	**TC-019**: not_run	—	covered
SUB-REQ-009	functional	**Supervisor as primary external interface point**	All command control status and logging communication between the RBNS and external systems shall be routed through the Procedure Supervisor with the exception of continuous sensor and imaging feeds (IF-EX-02a IF-EX-03 IF-EX-08) which connect directly to their consuming subsystem.	—	SYS-REQ-004	**TC-020**: not_run	—	covered
SUB-REQ-010	performance	**Lost-tracking alert within 100 ms**	The RBNS shall report a lost-tracking alert to the Procedure Supervisor within 100 ms of tracking loss detection.	—	SYS-REQ-006	**TC-021**: not_run	—	covered
SUB-REQ-011	safety	**Cease guidance commands within 50 ms of interlock deassert**	Upon deassertion of the motion enable signal (IF-EX-06a) the RBNS shall cease issuing guidance commands within 50 ms.	—	SYS-REQ-003	**TC-022**: not_run	—	covered
SUB-REQ-012	safety	**FAULT state and HOLD within 100 ms of lost-tracking**	Upon receiving a lost-tracking alert the Procedure Supervisor shall transition to the FAULT state and command HOLD on IF-EX-04 within 100 ms.	—	SYS-REQ-003	**TC-023**: not_run	—	covered

Req ID	Type	Title	Description	Source	Parent	Test Cases	Risk Controls	Coverage
SUB-REQ-013	functional	**Degraded pose estimate on single-source failure**	The RBNS shall maintain a degraded tip pose estimate upon failure of any single localization source and shall flag the estimate as DEGRADED on IF-03.	—	SYS-REQ-002	**TC-024**: not_run	—	covered
SUB-REQ-014	performance	**Path computation within 1 second**	The RBNS shall compute an airway path from current position to a newly commanded target within 1 second.	—	SYS-REQ-001	**TC-025**: not_run	—	covered
SUB-REQ-015	performance	**End-to-end guidance latency 200 ms**	The end-to-end latency from a tip pose estimate (IF-01) to the corresponding guidance command transmission (IF-EX-04) shall not exceed 200 ms.	—	SYS-REQ-001	**TC-026**: not_run	—	covered
SUB-REQ-016	functional	**CBCT trigger phase gating**	The RBNS shall issue CBCT acquisition triggers (IF-EX-02b) only during the NAVIGATE and CONFIRM procedure phases.	—	SYS-REQ-002	**TC-027**: not_run	—	covered
SUB-REQ-017	functional	**Procedure event logging**	The RBNS shall record all state transitions replanning events fault events and the tip pose trace to the Procedure Data Recorder (IF-EX-07) with timestamps of 10 ms resolution or better.	—	SYS-REQ-005	**TC-028**: not_run	—	covered
COMP-REQ-001	functional	**Tip articulation range (TBD)**	The Guidance Command Generator shall support distal tip articulation commands of up to [TBD] degrees per axis sufficient to follow airway centerline curvature at branch points up to the generation depth required by SYS-REQ-001.	—	SUB-REQ-014	**TC-029**: not_run	—	covered
COMP-REQ-002	functional	**Uncertainty ellipsoid bound (TBD)**	The Sensor Fusion and Uncertainty Estimator shall bound the fused position uncertainty ellipsoid such that the 95% confidence volume does not exceed [TBD].	—	SUB-REQ-002	**TC-030**: not_run	—	covered
COMP-REQ-003	functional	**Replanning threshold default (TBD)**	The Real-Time Path Updater shall default the replanning inhibition threshold to an uncertainty ellipsoid semi-axis of [TBD] configurable per SUB-REQ-005.	—	SUB-REQ-004	**TC-031**: not_run	—	covered